

How to Execute WordCount Program in MapReduce using Cloudera Distribution Hadoop(CDH)

For Lab on Wednesday (27/9/23)

The steps which show how to write a MapReduce code for Word Count.

Input:

Hello I am Geeks for Geeks

Hello I am an Intern

Output:

GeeksforGeeks 1

Hello 2

I 2

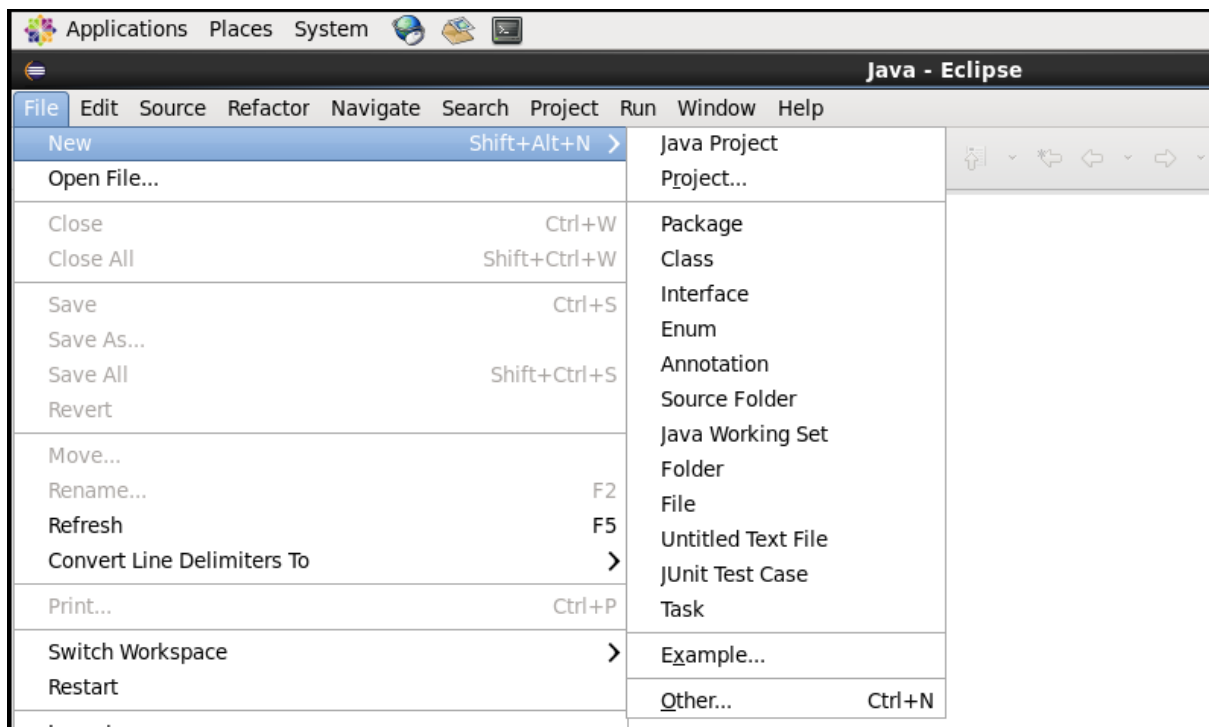
Intern 1

am 2

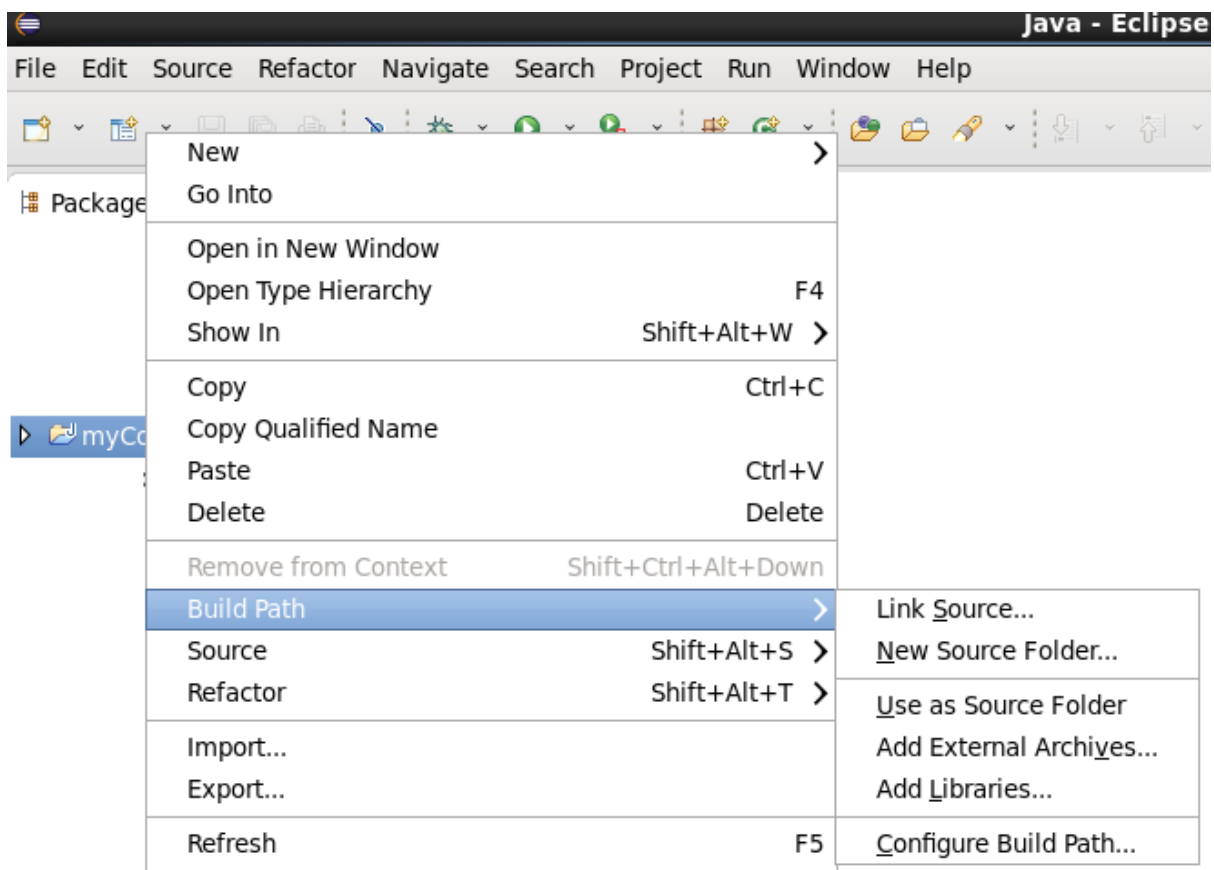
an 1

Steps:

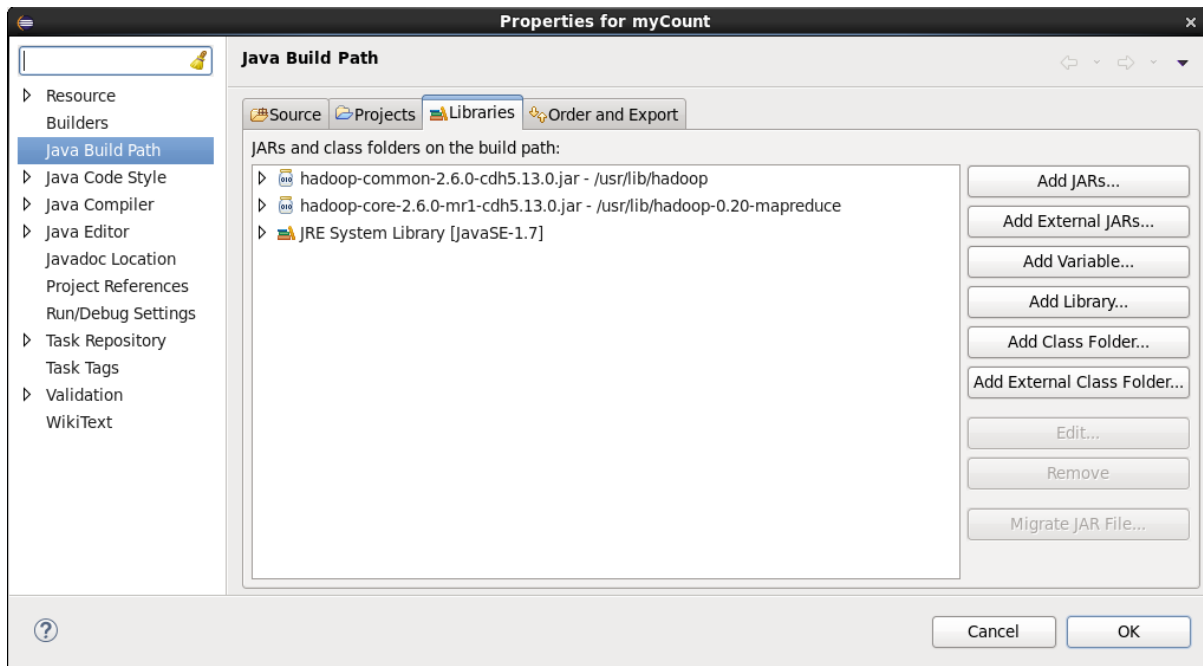
- First Open **Eclipse** -> then select **File** -> **New** -> **Java Project** ->Name it **WordCount** -> then **Finish**.



- Create Three Java Classes into the project. Name them **WCDriver**(having the main function), **WCMapper**, **WCREducer**.
- You have to include two Reference Libraries for that:
Right Click on **Project** -> then select **Build Path**-> Click on **Configure Build Path**



- In the above figure, you can see the Add External JARs option on the Right Hand Side. Click on it and add the below mention files. You can find these files in `/usr/lib/`
 1. `/usr/lib/hadoop-0.20-mapreduce/hadoop-core-2.6.0-mr1-cdh5.13.0.jar`
 2. `/usr/lib/hadoop/hadoop-common-2.6.0-cdh5.13.0.jar`



• **Mapper Code:** You have to copy paste this program into the WCMapper Java Class file.

- Java

```
// Importing libraries
import java.io.IOException;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;

public class WCMapper extends MapReduceBase implements Mapper<LongWritable,
                                                                    Text, Text, IntWritable> {

    // Map function
    public void map(LongWritable key, Text value, OutputCollector<Text,
```

```

        IntWritable> output, Reporter rep) throws IOException
    {

        String line = value.toString();

        // Splitting the line on spaces
        for (String word : line.split(" "))
        {
            if (word.length() > 0)
            {
                output.collect(new Text(word), new IntWritable(1));
            }
        }
    }
}

```

Reducer Code: You have to copy paste this program into the WCReducer Java Class file.

- Java

```

// Importing libraries
import java.io.IOException;
import java.util.Iterator;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;

public class WCReducer extends MapReduceBase implements Reducer<Text,
        IntWritable, Text, IntWritable> {

    // Reduce function
    public void reduce(Text key, Iterator<IntWritable> value,
        OutputCollector<Text, IntWritable> output,
        Reporter rep) throws IOException
    {

        int count = 0;

        // Counting the frequency of each words
        while (value.hasNext())
        {

```

```

        IntWritable i = value.next();
        count += i.get();
    }

    output.collect(key, new IntWritable(count));
}
}

```

Driver Code: You have to copy paste this program into the WCDriver Java Class file.

- Java

```

// Importing libraries
import java.io.IOException;
import org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;

public class WCDriver extends Configured implements Tool {

    public int run(String args[]) throws IOException
    {
        if (args.length < 2)
        {
            System.out.println("Please give valid inputs");
            return -1;
        }

        JobConf conf = new JobConf(WCDriver.class);
        FileInputFormat.setInputPaths(conf, new Path(args[0]));
        FileOutputFormat.setOutputPath(conf, new Path(args[1]));
        conf.setMapperClass(WCMapper.class);
        conf.setReducerClass(WCReducer.class);
        conf.setMapOutputKeyClass(Text.class);
        conf.setMapOutputValueClass(IntWritable.class);
        conf.setOutputKeyClass(Text.class);
        conf.setOutputValueClass(IntWritable.class);
    }
}

```

```

        JobClient.runJob(conf);
        return 0;
    }

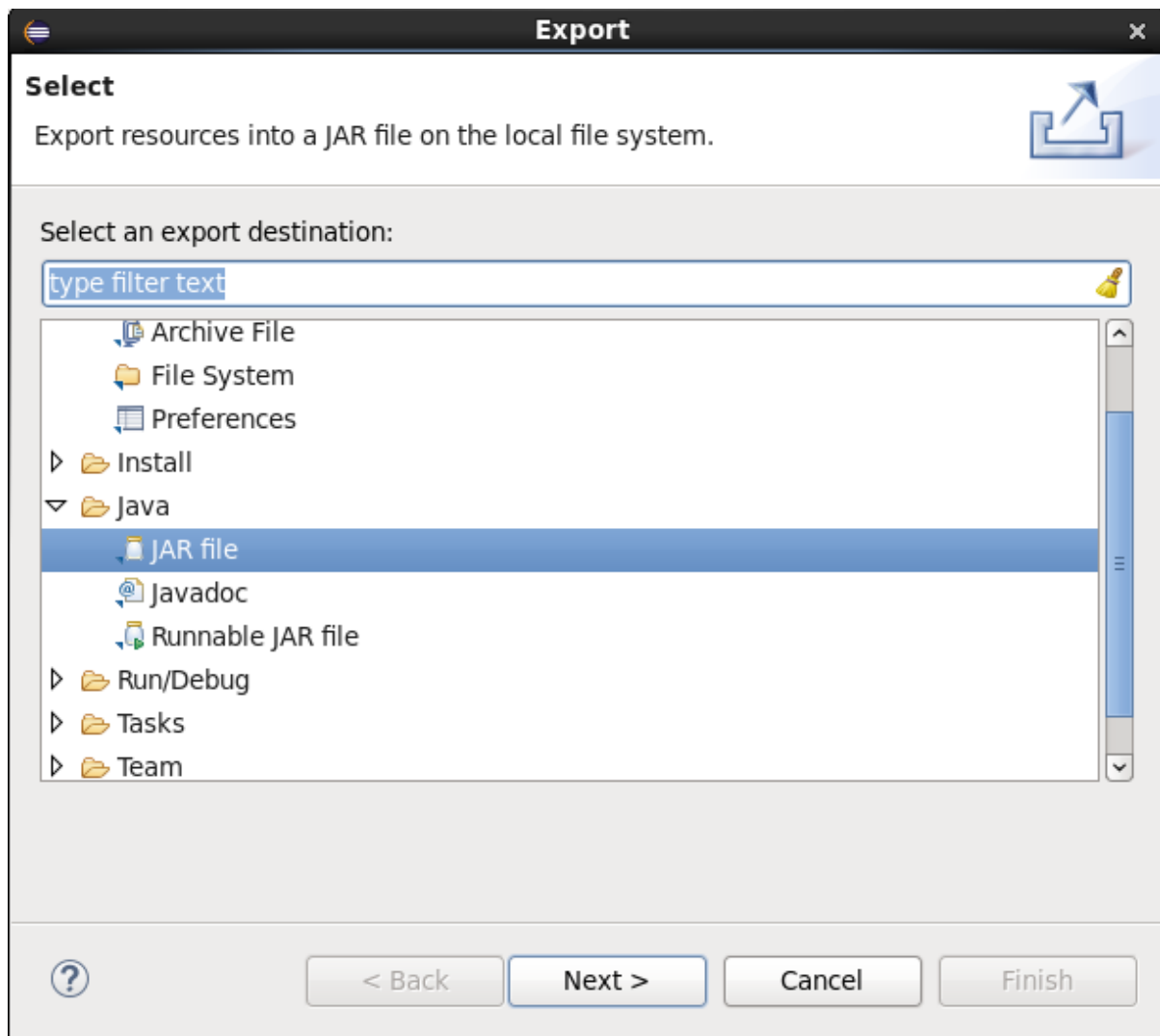
    // Main Method
    public static void main(String args[]) throws Exception
    {
        int exitCode = ToolRunner.run(new WCDriver(), args);
        System.out.println(exitCode);
    }
}

```

- Now you have to make a jar file. Right Click on **Project**-> **Click on Export**-> **Select export destination as Jar File**-> **Name the jar File**(WordCount.jar) -> **Click on next** -> at last **Click on Finish**. Now copy this file into the Workspace directory of Cloudera

New	>
Go Into	
Open in New Window	
Open Type Hierarchy	F4
Show In	Shift+Alt+W >
Copy	Ctrl+C
Copy Qualified Name	
Paste	Ctrl+V
Delete	Delete
Remove from Context	Shift+Ctrl+Alt+Down
Build Path	>
Source	Shift+Alt+S >
Refactor	Shift+Alt+T >
Import...	
Export...	
Refresh	F5
Close Project	

-



☒ Export generated class files and resources
☐ Export all output folders for checked projects
☐ Export Java source files and resources
☐ Export refactorings for checked projects. [Select refactorings...](#)

Select the export destination:

JAR file: ▼ Browse...

Options:

☒ Compress the contents of the JAR file
☐ Add directory entries
☐ Overwrite existing files without warning

?
< Back
Next >
Cancel
Finish

-
- Open the terminal on CDH and change the directory to the workspace. You can do this by using “cd workspace/” command. Now, Create a text file(**WCFile.txt**) and move it to HDFS. For that open terminal and write this code(remember you should be in the same directory as jar file you have created just now).
- **cat >> WordCCCountFinal.txt**
Enter your own text here
After finishing the text
Click Ctrl+z
- **Then Create a directory using below command:**
- **sudo -u hdfs hadoop dfs -mkdir /WordCCCount**
- **TO create the text file . Type the below command:**
- **Add the WordCCCount.txt in hadoop by using below command :**
- **sudo -u hdfs hadoop dfs -put**
/WordCCCount/WordCCCountFinal.txt
- **To view the contents**
- **sudo -u hdfs hadoop dfs -cat /WordCCCount/WordCCCountFinal.txt**
- **Run jar file now using below command :**

```
Sudo -u hdfs hadoop jar /home/cloudera/WordCCCountFinal.jar WCDriver  
/WordCCCount/WordCCCountFinal.txt OutputWC
```

- After executing the jar file , Run below command to see the output
- `hadoop fs -ls /WordCCCount`
- You can view output file that is OutputWC
- Type
- `sudo -u hdfs hadoop dfs -cat /WordCCCount/OutputWC`

NOTE: Don't follow these yellow highlighted commands

Thanks and Regards

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